

Watershed boundary extraction from digital elevation models using RBM-SegNet

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ABSTRACT

Accurately extracting watershed boundaries is critical for hydrological modeling and environmental management. Traditional extraction methods from Digital Elevation Models (DEMs) rely on manually defined thresholds and supplementary terrain features, limiting adaptability and efficiency. To address these issues, this study developed a watershed boundaries extraction framework based on a Residual Bottleneck Attention Multi-feature Fusion Network (RBM-SegNet). The framework consists of three components: an input layer, a semantic segmentation model, and a post-processing module. Key contributions include: (1) utilizing the [DEM, Slope, Hillshade, and Aspect] functions as the optimal input combination; (2) introducing residual connections and the Bottleneck Attention Module (BAM) to enhance feature transmission and suppress irrelevant regions; (3) incorporating multi-feature fusion to refine structural and detail prediction; and (4) incorporating post-processing to improve output-completeness and hydrological consistency. The experimental results show that RBM-SegNet outperforms traditional and existing deep learning methods in accuracy, demonstrating strong potential for practical applications.

1. Introduction

A fundamental unit in hydrological modeling and environmental management (Cho, 2025; Zhang et al., 2016; Liu et al., 2014; Tesfa et al., 2011), watersheds are areas delimited by ridges of high land that separate adjacent regions and drain precipitation into different river systems (Omernik et al., 2017; Lindsay et al., 2008). The accurate extraction of watershed boundaries, known as watershed delineation, is crucial for reliable hydrological analyses (Zhang et al., 2025; Choi and Engel, 2003).

Historically, watershed delineation was based on manually analyzing topographic maps, which relied heavily on expert interpretation and was impractical for large-scale applications (Karimipour et al., 2013; Sitanggang and Ismail, 2010). With the development of remote sensing and Geographic Information Systems, Digital Elevation Models (DEMs) have increasingly become the main data source for extracting watershed boundaries (Roy and Chowdhury, 2024; Xie et al., 2022; Rueda et al., 2013; O'Callaghan and Mark, 1984). Existing watershed boundary extraction methods based on DEMs can be classified as automatic methods based on surface-flow simulations and special methods

that consider complex terrain features. The automatic methods' workflows are relatively conformist, the key steps being to calculate the flow direction and determine the flow accumulation threshold (Bajjali, 2023; Xie et al., 2022; Jensen and Domingue, 1988). Existing methods typically employ the Deterministic 8 (D8) algorithm to establish the flow direction (Choi and Engel, 2003; Turcotte et al., 2001). For threshold setting, one approach is to assign a fixed threshold to the study area using trial-and-error or stream-density-based methods, by generating a series of simulated drainage networks and verifying whether their density and length align with actual hydrological patterns (Karmaker et al., 2022; Gopinath et al., 2014; Jensen and Domingue, 1988). To support large-scale applications, another approach adopts automated strategies that define tributaries based on a percentage increase in flow accumulation (Xie et al., 2022). Among these flow-routing-based techniques, the stream-burning method (Chen et al., 2024) has also been proposed to enhance the accuracy of drainage network extraction by locally lowering the elevations of DEM cells along predefined river channels. This preprocessing step effectively enforces the flow paths toward existing streams, thereby optimizing the simulated drainage network and potentially influencing the delineation of watershed boundaries. Although

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these methods perform well in most regions, they often fail to reproduce accurately the actual watershed boundaries in areas with complex terrain features — such as rivers, lakes, and polders — especially in flat terrain (Lai et al., 2016). To improve accuracy, some studies have developed watershed boundary extraction methods that incorporate multiple terrain features, enhancing the connectivity of the drainage network and extracting more reasonable boundary results (Zhang et al., 2025; Li et al., 2019; Lai et al., 2016). However, both method categories depend heavily on threshold parameter settings, and the latter require the integration of diverse topographic datasets, which imposes strict preprocessing requirements and increases methodological complexity.

With the rapid development of deep learning, its uses in terrain analysis have expanded significantly (Uhe et al., 2025; Du et al., 2024; Xu et al., 2021; Li et al., 2019). These methods can learn features directly from input data, reducing the manual intervention (Jaroenchai et al., 2024). With pixel-level classification capability, semantic segmentation performs well in identifying hydrological features with strong spatial continuity. In recent years, semantic segmentation has been increasingly applied to key steps in traditional watershed boundary-extraction workflows, such as for drainage network extraction and terrain feature identification. For example, Jaroenchai et al. (2024) combined topographic position index and slope aspect information to enhance drainage network extraction under various terrain conditions. Additionally, Mao et al. (2021) proposed a transfer learning method based on fine-tuning deep convolutional neural networks, combining models pre-trained on images to enhance the accuracy and transferability of streamline delineation. Li et al. (2021) developed a robust water extraction network using an improved U-Net model trained on Sentinel-1 and Sentinel-2 remote-sensing images. Although these approaches often generate noisy outputs and face challenges in capturing boundaries structures in regions with smooth or ambiguous transitions, their work demonstrates the potential of semantic segmentation for non-water feature extraction from DEMs.

Therefore, this study approaches watershed boundary extraction as a semantic segmentation task, eliminating the need to define thresholds manually or integrate various terrain features. However, the baseline model, SegNet (Badrinarayanan et al., 2017), has limitations in segmenting narrow boundaries or modeling boundary continuity. To address this, this study developed an improved deep learning model, RBM-SegNet. The model alleviates information loss during deep feature transmission through residual networks (He et al., 2016) and enhances the learning and expression of key information using a Bottleneck Attention Module (BAM) (Park et al., 2018). Meanwhile, integrating a multi-feature fusion strategy enhances semantic perception and detail restoration at each decoder layer, further improving the model's segmentation accuracy. Considering the characteristics of DEMs, various input combinations were evaluated, and a four-channel input [DEM, Slope, HillShade, Aspect] was adopted to enhance the model's ability to perceive terrain features. Compared with other deep learning models, the proposed method demonstrated superior performance, highlighting its effectiveness in learning spatial features. Furthermore, by incorporating post-processing techniques with flow accumulation, the outputs' connectivity was further optimized, producing final outputs that more closely align with natural hydrological structures. Compared with the traditional D8 flow-routing method, the proposed method exhibits adaptability across diverse terrain conditions, further validating its ability to extract watershed boundary features accurately. The results indicate that semantic segmentation models offer a direct and efficient solution for watershed boundary extraction tasks.

2. Datasets and preprocessing

As Fig. 1 shows, we used the National Hydrography Dataset Plus (NHDPlus) High Resolution (with 10-m spacing) for the contiguous United States (U.S. Geological Survey, 2019) to train and initially evaluate the proposed framework. From this dataset, the 3D Elevation

Program (3DEP) (Snyder, 2018) raster data (Fig. 1(a)), resampled to a vertical resolution of 60 m, were mainly used. We selected 60 m because it reduces the fine-scale undulations and shallow depressions retained at 30 m while avoiding the over-smoothing observed at 90 m, thereby yielding a more continuous and consistent watershed-boundary delineation. Additionally, vector data from the Water Boundary Vector Dataset (WBD) were rasterized and used as watershed mask labels.

For model training, we randomly cropped 46 large regions from the study area, with 90% allocated for training and validation, and the remaining 10% for testing. Given the size of the cropped areas, a sliding window approach was employed to divide them into patches of 512×512 pixels. To enhance the model's generalization capability, the training and validation data were augmented by rotating the images by 90° , 180° , and 270° and applying horizontal flipping. In total, 80% (2952 samples) of the cropped patches were randomly selected for the training set, and 20% (738 samples) for the validation set.

The patch size of 512×512 pixels was chosen as a trade-off between spatial context and computational efficiency. Smaller patches (e.g., 256×256) tend to lose large-scale terrain context, while larger patches (e.g., 1024×1024) increase computational burden without improving segmentation accuracy. Hence, 512×512 was selected as an empirically balanced size for stable training and accurate watershed boundary delineation.

To evaluate further how well the model's generalized across regions, we adopted the Copernicus DEM (COP-DEM) dataset from the European Space Agency (ESA) as the primary elevation data for our study area, which is located in China (Fig. 2). This dataset was also resampled to a vertical resolution of 60 m and used for cross-regional performance evaluation.

2.1. Topographic features

A single DEM has limited representational capability. To improve the segmentation accuracy, this study adopted a multi-channel input design. Topographic features, including slope, hillshade, and aspect, were derived from the DEM using the RichDEM Tools (Barnes, 2016) and Relief Visualization Toolbox (Kokalj and Somrak, 2019). Previous studies have extensively used slope and aspect for extracting drainage networks (Jaroenchai et al., 2024; Mao et al., 2021). Slope represents the degree of inclination of the surface relative to the horizontal plane (Nair et al., 2017), while aspect refers to the slope direction (Raaflaub and Collins, 2006). Hillshade generates a 3D terrain representation using an illumination source (Khan and Kumar, 2024), intended to visualize the surface relief better, similar to interpreting remote sensing imagery (Van Den Eeckhaut et al., 2005).

2.2. HydroSHEDS reference

HydroSHEDS provides global-scale hydrological features and has been widely applied in previous research (Liu et al., 2020; Gao et al., 2018). For this study, we retrieved the HydroSHEDS watershed data (as_bas_15s) with an approximate spatial resolution of 410 m from the USGS website (<http://hydrosheds.cr.usgs.gov>). We selected the highest available level of sub-basin segmentation, which has an average area of approximately 130.6 km². These data were used as reference boundaries to evaluate the watershed extraction results based on the COP-DEM dataset (Fig. 2(c)).

3. Methods

3.1. Deep learning network architecture

This study developed and tested an improved deep neural network, RBM-SegNet, based on the classic SegNet (Badrinarayanan et al., 2017). The network maintains SegNet's symmetrical encoder-decoder design

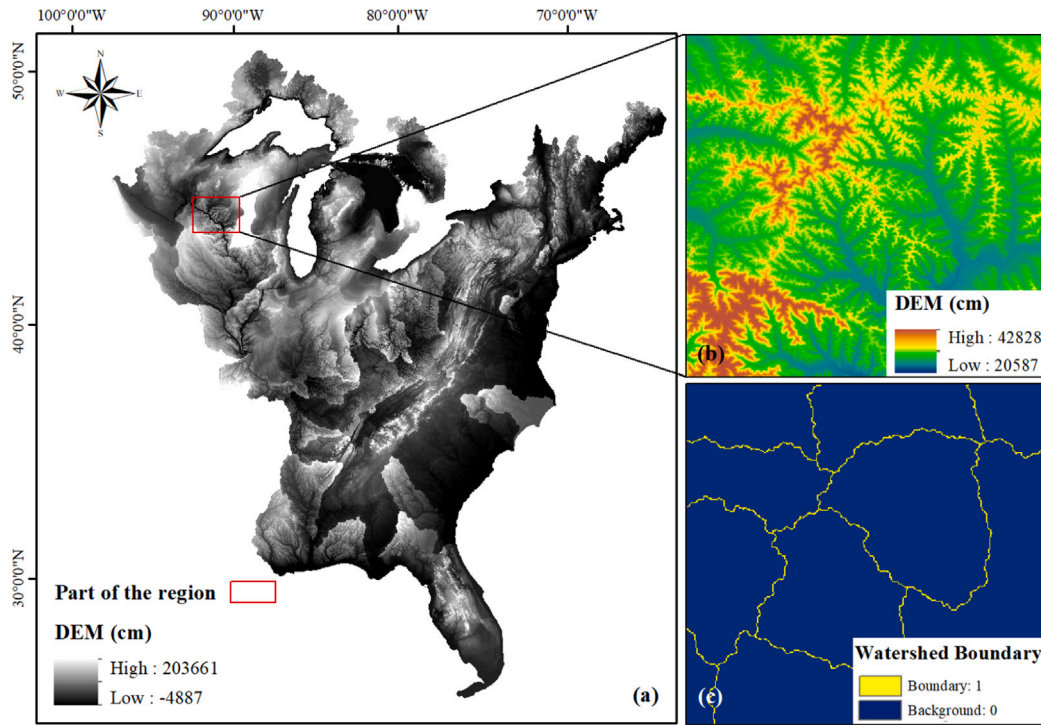


Fig. 1. The National Hydrography Dataset Plus (NHDPlus) High Resolution (U.S. Geological Survey, 2019) used in this study: (a) and (b) are the 3DEP digital elevation model; (c) is the water boundary vector data; (b)–(c) are inside Wisconsin, USA.

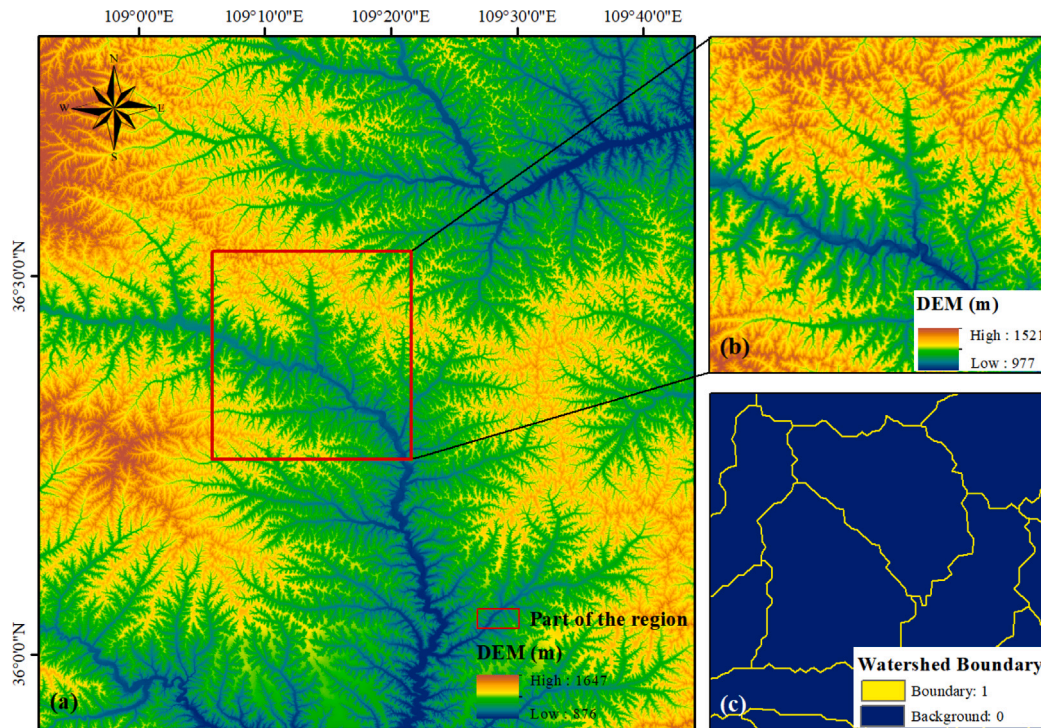


Fig. 2. The COP-DEM data used in this study: (a) and (b) are the digital elevation model; (c) is the HydroSHEDS watershed data (as_bas_15s) at approximately 410 m resolution; (a)–(c) are inside Shannxi, China.

while introducing enhancement mechanisms in several key modules to improve the model's capability for semantic feature representation.

SegNet is a pixel-level image segmentation network, characterized by a symmetrical encoder–decoder architecture. The encoder, typically based on the VGG16 backbone, extracts deep features from the input image through multiple convolutional layers and max-pooling

operations, while recording pooling indices for use in the subsequent nonlinear upsampling during decoding. The decoder utilizes these pooling indices to restore spatial resolution, thereby avoiding the blurring effects caused by interpolation, and progressively reconstructs the spatial structure of the feature maps with the aid of convolutional layers. This design significantly reduces the number of model parameters and

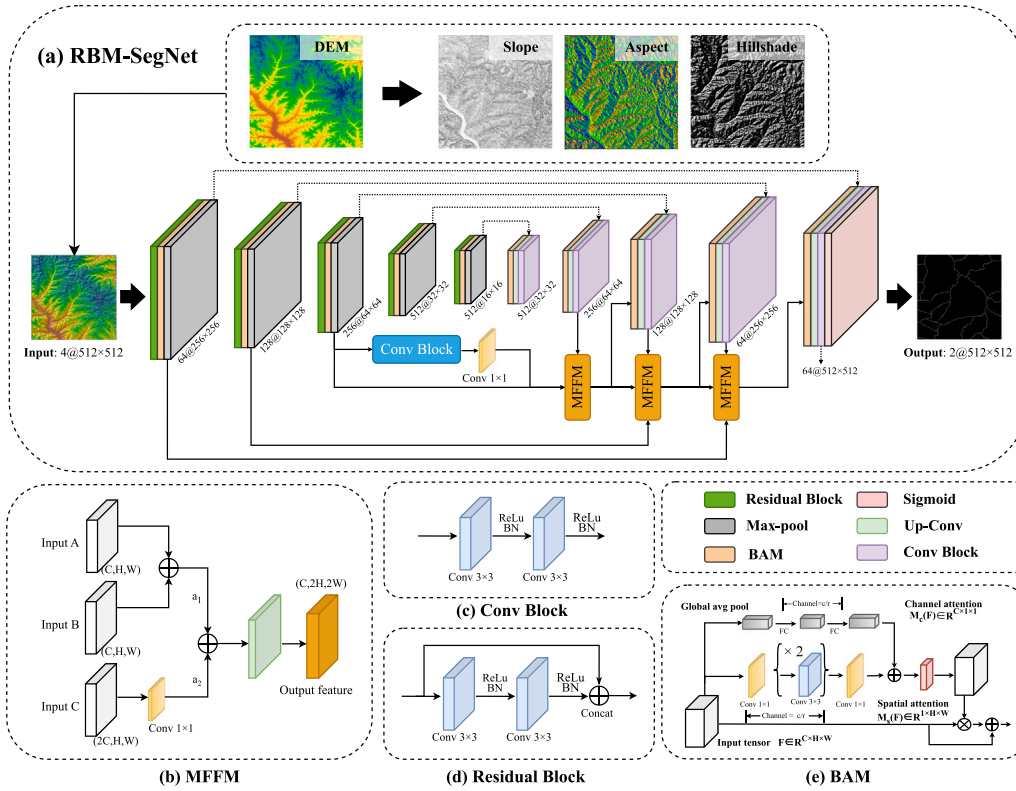


Fig. 3. The structure diagram of RBM-SegNet.

improves computational efficiency, which constitutes a key advantage of SegNet.

However, the original model faces several limitations when segmenting elongated structures (Jiang and Jiang, 2024). The encoder uses fixed-size convolutional kernels, resulting in a limited receptive field that can only capture local elevation variations. In contrast, watershed boundaries often manifest as elongated, linear structures that follow continuous terrain trends and typically lack prominent local edge features. This makes it difficult for convolutional networks to model the global spatial relationships required for accurate segmentation. Furthermore, as the network deepens, critical fine-grained information in the lower layers — such as abrupt changes in elevation or slope — is progressively compressed through repeated downsampling, thereby weakening the network’s ability to represent complex boundary details. Conversely, the SegNet decoder relies solely on pooling indices from the encoder for layer-by-layer upsampling. Although this approach preserves some spatial location information, the strictly unidirectional reconstruction process limits contextual awareness, particularly in geographically complex regions. Consequently, the decoder struggles to recover well-defined boundaries accurately in areas where topographic transitions are smooth or ambiguous.

To address the aforementioned issues, the overall architecture of the proposed RBM-SegNet is illustrated in Fig. 3(a). The network takes four-channel inputs and adopts ResNet-34 (He et al., 2016) as the encoder backbone, where residual connections are introduced to reduce feature loss effectively. Additionally, the BAM (Park et al., 2018) enhances the model’s ability to learn and express key information. As downsampling progresses, the feature maps’ spatial resolution gradually decreases to $1/32$, while the number of channels increases to capture deeper semantic information. In the decoding phase, each decoder layer consists of BAM, an upsampling layer, and 3×3 convolution layers, which progressively restore spatial details and optimize feature representation. To compensate for information loss caused by upsampling, the Multi-Features Fusion Module (MFFM) was proposed and added to the 2nd, 3rd, and 4th upsampling stages of the decoder, which fully fuses

the semantic information from both the downsampling and upsampling layers, resulting in predictions with clearer structures and finer details. Finally, to address the class imbalance in pixel-wise watershed, RBM-SegNet uses a hybrid Dice Loss and Binary Cross-Entropy (BCE) Loss. The sections that follow detail the MFFM, BAM, and loss functions.

3.1.1. Multi features fusion block

In the SegNet encoder-decoder architecture, as the network depth increases, the spatial resolution of feature maps gradually decreases while the semantic information richens. During the decoding phase, although upsampling using pooling indices helps recover spatial structures, the absence of direct skip connections often causes the loss of semantic information during reconstruction. To address this limitation, this study introduces a Multi-Feature Fusion Module (MFFM), which effectively integrates features from different encoder and decoder levels, thereby mitigating the restricted information flow inherent in SegNet. As Fig. 3(b) shows, the MFFM fuses the encoder features, decoder features, and the output from the previous fusion module at the current level, establishing a cross-level pathway for enhanced feature representation.

Specifically, let the encoder feature map at the current decoding stage be denoted as $E(A)$, the decoder output at the same stage as $D(B)$, and the output from the previous MFFM module as $M(C)$. The fusion process is formulated as follows:

$$F_{fusion} = a_1 \times (E(A) + D(B)) + a_2 \times (\Phi(M(C))) \quad (1)$$

where a_1 and a_2 are fusion coefficients used to balance the contributions of the current layer’s features and the cross-level features. Based on experimental tuning, their optimal values are set to $a_1 = 0.7$ and $a_2 = 0.3$. ϕ represents a 1×1 convolution operation applied to the output of the previous MFFM to reduce channel dimensionality and enhance feature representation, thereby improving non-linear expressiveness. To ensure proper alignment, all resized feature maps undergoing fusion must match regarding channel dimensions and spatial resolution. The resulting fused feature map F_{fusion} is then fed into the next decoding layer as input.

3.1.2. Bottleneck Attention Module (BAM)

To enhance the model's ability to focus on critical features, this paper introduces the BAM (Park et al., 2018) illustrated in Fig. 3(e). By combining channel attention and spatial attention mechanisms, the module improves the model's sensitivity to salient regions.

The channel attention module is designed to evaluate the relative importance of each channel in the overall semantic representation. Given an input feature map $F = R^{C \times H \times W}$, global average pooling is first applied to extract contextual information from each channel, resulting in a channel descriptor vector $F = R^{C \times 1 \times 1}$. This vector is then fed into a multilayer perceptron with a single hidden layer to model the inter-channel dependencies. To reduce the computational cost, the hidden layer dimension is compressed to $\frac{C}{r}$, where r denotes the reduction ratio. The resulting channel attention weights lie in the range $[0, 1]$ and indicate the relative importance of each channel. These weights are broadcast across all spatial positions of the input feature map and multiplied channel-wise to enhance the informative features while suppressing redundant ones. The spatial attention module is designed to highlight discriminative spatial regions within the feature map. First, a 1×1 convolution is applied to the input feature map F to reduce the number of channels and decrease computational complexity. Then, two successive 3×3 dilated convolutions are employed to enlarge the receptive field and capture richer contextual information. Finally, another 1×1 convolution compresses the feature map into a single-channel spatial attention map.

The final attention map $M(F)$ is obtained by combining the channel attention and spatial attention maps through element-wise addition, followed by normalization using a Sigmoid function:

$$M(F) = \sigma(M_c(F) + M_s(F)) \quad (2)$$

where σ denotes the Sigmoid activation function. This attention weight is then applied to the input feature map F to produce the final weighted feature map:

$$F_{fusion} = M(F) \otimes F + F \quad (3)$$

3.1.3. Loss-function

Due to the sparsity of watershed boundaries and the limited number of labeled pixels, the positive and negative samples are severely imbalanced. Therefore, this study appropriately adjusts the weights in the BCE Loss function for the model to focus more attention on the less frequent samples, thereby balancing the sample distribution. Since the Dice loss function is often used to solve imbalance problems, this study combines Dice loss with BCE loss, formulated as follows:

$$BCELoss = - \sum_{i=1}^n [y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i)] \quad (4)$$

$$DiceLoss = 1 - \frac{2 \sum_{i=1}^n \hat{y}_i y_i}{\sum_{i=1}^n \hat{y}_i + \sum_{i=1}^n y_i} \quad (5)$$

$$Loss = \sigma \times BCELoss + (1 - \sigma) \times DiceLoss \quad (6)$$

where n is the total number of pixels in the image, y_i represents the pixel label value, $\hat{y}_i$ is the predicted value, and σ is an adjustable parameter used to balance the loss function's foreground focus. All experiments in this study were conducted under the condition of $\sigma = 0.7$.

3.2. Postprocessing for extracting boundaries

The semantic segmentation model obtained an initial prediction of watershed boundaries represented as a boundary probability map, where each pixel value indicated the softmax probability of its corresponding class. To refine the boundary prediction, we first thresholded the map and used a Python skeleton function to extract the centerline of the boundary, creating the watershed boundary skeleton. To address

open boundaries, we adopted a two-step preprocessing strategy. For slight gaps, morphological closing operations in Python were used to repair minor discontinuities. For larger breaks, we incorporated flow accumulation (FA) information to guide the correction. Specifically, we identified intersection points between the predicted watershed boundaries and the FA data, and then performed reverse tracing along the flow direction based on the D8 algorithm. If the traced paths flowed into different intersection points, they were classified as belonging to different watersheds. Subsequently, starting from the endpoints of broken boundary segments, we traced along the boundary region between adjacent watershed labels to fill in the missing parts and ensure the continuity of the watershed boundary. To further improve boundary accuracy, we examined the 8-neighborhood of each boundary pixel: if multiple watershed labels were present, the pixel was considered a valid boundary location; otherwise, it was replaced by the neighboring pixel with the highest elevation. Finally, to remove noise and artifacts caused by misclassification or fragmented segments, we discarded boundary skeleton segments with a length ≤ 30 pixels. The FA data were generated using the Pysheds library (Bartos, 2020) based on the D8 algorithm.

3.3. Evaluation metrics

This study employed four commonly used metrics to evaluate the proposed model's performance: Accuracy, F1-score, True Negative Rate (TNR), Mean Intersection over Union (mIoU), and Frequency Weighted Intersection over Union (FWIoU).

Accuracy measures the proportion of all samples correctly predicted, as calculated using Eq. (7):

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (7)$$

where TP denotes true positives, FP denotes false positives, FN denotes false negatives, and TN denotes True negatives. The F1-score is the harmonic mean of precision and recall, which considers both false positives and false negatives. It is particularly effective in evaluating model performance on imbalanced classification tasks, as calculated using Eq. (8):

$$F1\text{-Score} = \frac{2 \times (\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}} \quad (8)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (9)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (10)$$

TNR measures a model's ability to correctly identify negative instances and reflects its control over false positives. In this task, the positive class is boundary pixels, and the negative class is non-boundary pixels; a higher TNR indicates cleaner background predictions with fewer spurious boundaries. The metric is defined in Eq. (11):

$$TNR = \frac{TN}{TN + FP} \quad (11)$$

where TN denotes the number of true negatives and FP denotes the number of false positives. mIoU evaluates segmentation performance by calculating the average ratio between the intersection and the union of predicted and ground truth labels across all classes, as presented in Eq. (13):

$$IoU = \frac{TP}{TP + FP + FN} \quad (12)$$

$$mIoU = \frac{1}{N} \sum_{i=1}^N IoU_i \quad (13)$$

where N denotes the total number of classes. FWIoU is an extension of mIoU that assigns a weight to each class based on its frequency in the dataset. This metric is especially suitable for handling imbalanced

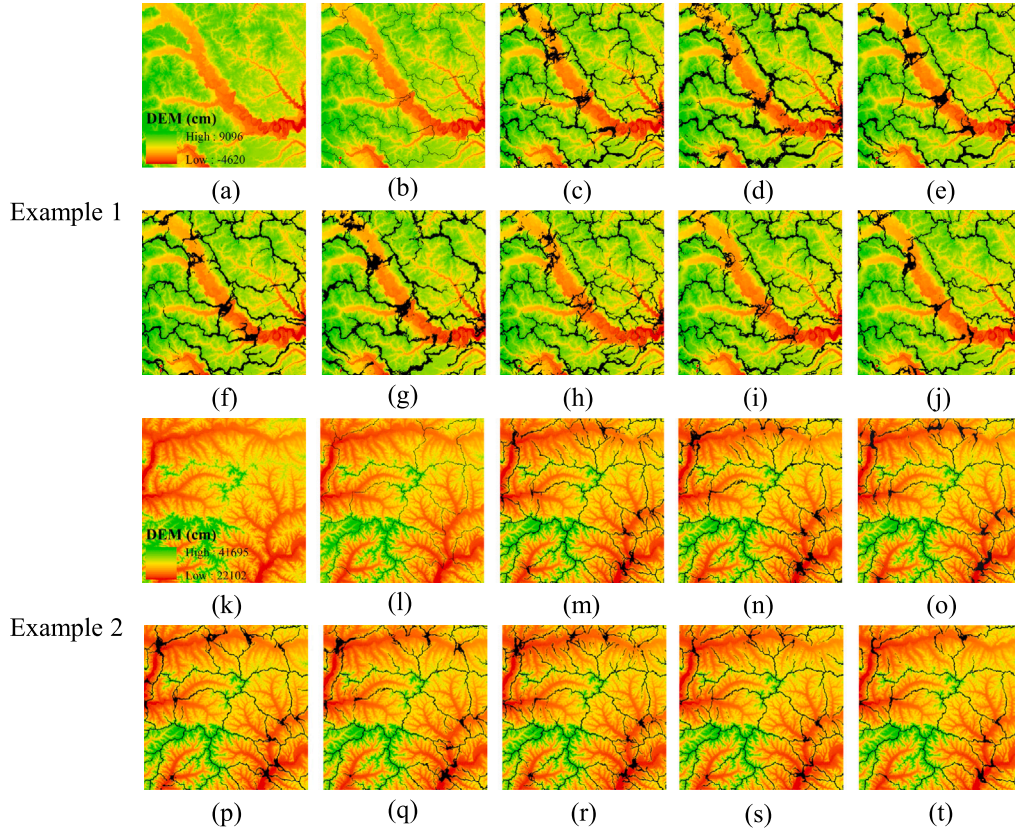


Fig. 4. Watershed boundary extraction results for two example areas using RBM-SegNet with different input combinations, where black lines are the watershed boundaries. (a) and (k): Original DEM; (b) and (l): Ground truth boundaries; (c) and (m): Result using DEM only; (d) and (n): DEM + Slope; (e) and (o): DEM + Hillshade; (f) and (p): DEM + Aspect; (g) and (q): DEM + Slope + Hillshade; (h) and (r): DEM + Slope + Aspect; (i) and (s): DEM + Aspect + Hillshade; (j) and (t): DEM + Slope + Aspect + Hillshade.

class distributions and provides a more representative evaluation of the model's overall segmentation performance, as presented in Eq. (14):

$$FWIoU = \frac{\sum_{i=1}^N IoU_i \times N_i}{\sum_{i=1}^N N_i} \quad (14)$$

where N_i denotes the pixel count for class i .

3.4. Experiment settings

We used the Adam optimizer with an initial learning rate of 0.00001. The batch size was set to 8, and the maximum number of epochs was 100. To obtain a stable model, we implemented an adaptive convergence mechanism. Training was stopped early if the validation set loss did not improve by more than 0.1 over 10 consecutive rounds. Additionally, the learning rate was reduced by 20% if the training set loss exceeded the minimum recorded loss, and training was resumed from the previous round.

All experiments were conducted on an NVIDIA GeForce RTX 3090 with 24 GB of VRAM, running on Ubuntu 18.04 LTS. The model was built using Python 3.8.2 with PyTorch 1.7.2 as the backend. The Python libraries used include scikit-image 0.18.1, scikit-learn 0.24.0, Rtree 0.9.4, GDAL 3.0.2, and NumPy 1.19.4. The software environment was managed via Anaconda 4.10.0.

4. Results and discussion

4.1. Evaluating the proposed framework's input layer

This section reports on the impact of different topographic feature combinations on the watershed boundary extraction task, demonstrating the necessity of using a four-channel input [DEM, Slope, Aspect,

Hillshade]. Under the same experimental conditions and dataset, we trained the RBM-SegNet model with various topographic feature combinations and compared their performance. Table 1 and Fig. 4 present the results of extracting watershed boundaries using these combinations.

The experimental results show that in the dual-channel settings, the combination with only Hillshade features performed the best. Hillshade captures the boundaries between different mountain slopes through assumed illumination conditions, which is particularly important for adapting to terrain changes in complex environments. Furthermore, the accuracy of watershed boundary extraction significantly improves when combined with Slope and Aspect information. Slope helps the model accurately identify the protrusions of ridgelines, while by describing the direction the terrain faces, aspect provides a strong supplement to the extraction of Hillshade features.

Ultimately, the four-channel input [DEM, Slope, Aspect, Hillshade] performed the best in the watershed boundary extraction task. Data input 8, along with the four-channel input, outperformed other settings across all evaluation metrics and showed significant improvement in avoiding misclassifications, as Fig. 4 shows.

4.2. Evaluating the semantic segmentation models for watershed boundary extraction

This section reports the performance of different semantic segmentation models in the task of watershed boundary extraction. We compared RBM-SegNet with various semantic segmentation models on the same test dataset. Table 2 and Fig. 5 present the models' performance metrics and boundary extraction results.

Regarding model selection, SegNet (Badrinarayanan et al., 2017) was chosen as the baseline model, resulting from comprehensively

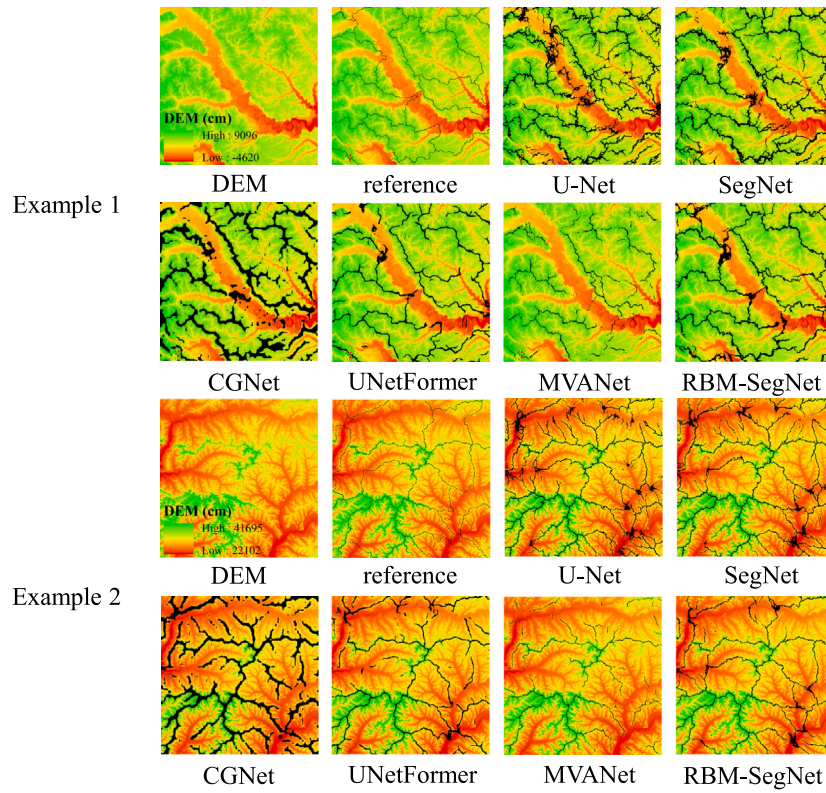


Fig. 5. Comparison of watershed boundaries extraction using U-Net, SegNet, CGNet, UNetFormer, MVANet and RBM-SegNet for two example areas, where black lines are the watershed boundaries.

Table 1

Accuracy assessment of initial watershed boundaries extraction using RBM-SegNet. Bold fonts represent the highest Accuracy, F1-score, mIoU and FWIoU achieved from RBM-SegNet, with inclusion of topographic features.

Data input	Topographic features				Accuracy	F1-score	mIoU	FWIoU	TNR
	DEM	Slope	HillShade	Aspect					
1	✓				90.90%	64.23%	55.67%	89.09%	91.22%
2	✓	✓			87.76%	59.60%	51.78%	85.89%	88.09%
3	✓		✓		92.08%	64.35%	56.15%	90.29%	92.53%
4	✓			✓	90.50%	63.12%	54.76%	88.67%	90.46%
5	✓	✓	✓		88.64%	61.00%	52.80%	86.78%	89.61%
6	✓	✓		✓	88.66%	61.39%	52.93%	86.79%	88.94%
7	✓		✓	✓	91.01%	63.58%	55.21%	89.19%	92.45%
8	✓	✓	✓	✓	92.33%	64.46%	56.26%	90.52%	94.75%

Table 2

Accuracy assessment of initial watershed boundary extraction using U-Net, SegNet, CGNet, UNetFormer, MVANet and RBM-SegNet.

Models	Accuracy	F1-score	mIoU	FWIoU	TNR	Parameters
U-Net (Ronneberger et al., 2015)	88.41%	60.34%	52.08%	86.51%	88.62%	19.65M
SegNet (Badrinarayanan et al., 2017)	90.74%	62.14%	54.04%	88.88%	91.15%	18.82M
CGNet (Wu et al., 2020)	81.49%	52.04%	44.56%	79.48%	81.92%	0.49M
UNetFormer (Wang et al., 2022)	90.71%	62.13%	53.43%	88.05%	90.66%	11.72M
MVANet (Yu et al., 2024)	97.26%	63.11%	56.06%	95.34%	94.51%	94.14M
RBM-SegNet	92.33%	64.46%	56.26%	90.52%	94.75%	23.29M

evaluating multiple models in preliminary experiments. Although some lightweight models have advantages in terms of parameter efficiency, their accuracy is somewhat limited. Conversely, certain larger models showed slight improvements in accuracy, their excessive parameter sizes and computational costs limited their practical deployment. In contrast, having only approximately 20% of the parameters of Swin Transformer-based models, RBM-SegNet achieved comparable or even superior accuracy, demonstrating higher computational efficiency and application potential.

From the quantitative results, RBM-SegNet outperformed SegNet in several key metrics, including Accuracy (+1.59%), F1-score (+2.32%), mIoU (+2.22%), FWIoU (+1.64%) and TNR (+3.60%). Particularly in metrics measuring segmentation capability, such as mIoU, F1-score and TNR, RBM-SegNet achieved the highest values (56.26, 64.46 and 94.75, respectively), indicating its robust ability to identify boundary locations accurately. It is worth noting that FWIoU is weighted by class frequency; therefore, when non-boundary pixels dominate, even a slight increase in background IoU is amplified into a substantial gain

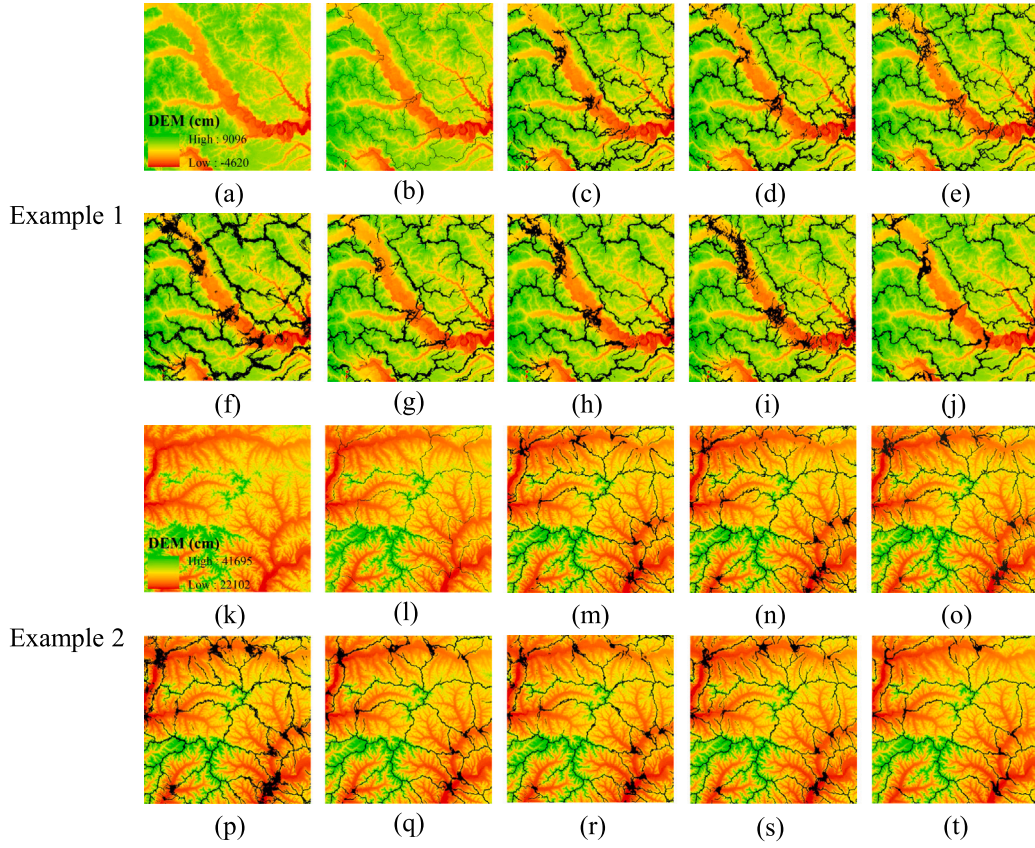


Fig. 6. Watershed boundary extraction results for two example areas using RBM-SegNet with different modules, where black lines are the watershed boundaries. (a) and (k): Original DEM; (b) and (l): Ground truth boundaries; (c) and (m): Result using SegNet; (d) and (n): SegNet + Residual; (e) and (o): SegNet + BAM; (f) and (p): SegNet + MFFM; (g) and (q): SegNet + Residual + BAM; (h) and (r): SegNet + Residual + MFFM; (i) and (s): SegNet + BAM + MFFM; (j) and (t): RBM-SegNet.

in FWIoU. MVANet makes more conservative negative predictions and achieves a slightly higher background IoU, hence a higher FWIoU. In contrast, RBM-SegNet emphasizes boundary depiction and thus excels in F1-score and mIoU; however, the accompanying small increase in false positives lowers its overall FWIoU after frequency weighting.

To illustrate the extraction performance, Fig. 5 presents the results of each model. RBM-SegNet effectively enhances the ability to extract boundary features in complex backgrounds by introducing the residual connection architecture and BAM. By combining MFFM, it effectively mitigates the information loss during upsampling and enhances the model's ability to predict clearer structures and finer details. Compared to other models, the boundaries extracted by RBM-SegNet exhibited greater coherence and accuracy. Especially in regions with complex terrain, such as wide river channels, the extracted boundaries aligned more closely with the actual topographic morphology, with significantly reduced error areas and improved detail retention. In contrast, U-Net (Ronneberger et al., 2015) could roughly extract the watershed boundaries on a large scale, but due to the loss of some detailed features during upsampling, the boundary lines appeared discontinuous and high-frequency artifacts. Although CGNet (Wu et al., 2020) adopts a lightweight architecture that provides advantages in computational efficiency, its limited network depth and receptive field, along with weak feature fusion capability, make the model prone to misclassification and local pseudo-boundaries. The overall boundary continuity and closure are poor, resulting in a relatively low mIoU of only 44.56%. UNetFormer (Wang et al., 2022) significantly enhances global spatial modeling capability by introducing a Transformer-based encoder. However, since its decoder still relies on upsampling to reconstruct spatial

details, slight blurring and discontinuities remain in local edge regions. Incorporating multi-view learning and a cross-attention mechanism, MVANet (Yu et al., 2024) achieved high overall pixel classification accuracy and produced smoother watershed boundaries. However, visible discontinuities in local areas suggest limitations in capturing complete boundaries.

In summary, RBM-SegNet significantly improved the accuracy and completeness of watershed boundary extraction. Although its parameter count increases by approximately 4M compared to the baseline model, it remains considerably more compact than large-scale models, achieving a better performance–complexity trade-off, providing an efficient and reliable solution for watershed boundary extraction tasks. However, certain artifacts can still be observed in the raw segmentation results. These artifacts mainly include overly thick boundaries, discontinuities, small-scale fractures, and isolated boundary fragments. Their occurrence can be attributed to two main factors. First, in transition zones, the spatial scale of feature variation is limited and elevation gradients are not pronounced, making it difficult for the model to accurately distinguish pixels. This often leads to local discontinuities, fractures, or isolated segments along the boundaries. Second, during deep feature fusion, convolution and upsampling operations tend to cause boundary blurring, resulting in classification uncertainty among adjacent pixels and consequently over-smoothed or overly wide boundary representations. To enhance the accuracy of the extracted boundaries, a post-processing procedure is applied to the initial segmentation results. This includes boundary refinement, removal of non-closed boundaries, and connectivity enhancement based on flow accumulation (FA). The overall workflow of this refinement process is illustrated in Fig. 7.

Table 3
Ablation study results of RBM-SegNet under different module configurations.

Model SegNet	Module			Accuracy	F1-score	mIoU	FWIoU	TNR
	Residual	BAM	MFFM					
1				90.74%	62.14%	54.04%	88.88%	90.92%
2	✓			91.07%	62.60%	54.47%	88.97%	91.00%
3		✓		91.44%	62.65%	54.87%	89.55%	91.75%
4			✓	90.87%	62.88%	54.71%	88.90%	90.97%
5	✓	✓		91.28%	63.27%	55.21%	89.64%	91.85%
6	✓		✓	91.98%	63.42%	55.38%	89.99%	92.27%
7		✓	✓	92.15%	63.97%	55.91%	90.12%	92.54%
8	✓	✓	✓	92.33%	64.46%	56.26%	90.52%	94.75%

4.3. Evaluation of the residual, BAM, and MFFM modules through ablation experiments

This section conducted comparative experiments on the original SegNet, SegNet with added Residual Block, BAM, MFFM, and the final integrated RBM-SegNet model. Multiple evaluation metrics, including Accuracy, F1-score, mIoU, FWIoU, and TNR, were used to assess the influence of each module on model performance. The experimental results are summarized in Table 3, and visual comparisons are presented in Fig. 6. A step-by-step comparison allows a clear analysis of the contribution and mechanism of each component.

As shown in Table 3, the baseline SegNet demonstrates stable performance across all metrics, providing a solid foundation for subsequent improvements. After introducing the Residual Block, all performance indicators show consistent improvement, with the F1-score increasing to 62.60% and mIoU to 54.47%. This demonstrates that the module effectively reduces feature loss during forward propagation. By introducing shortcut connections, the Residual structure alleviates the vanishing-gradient problem in deep networks, enabling efficient learning of meaningful features at greater depths. As illustrated in Fig. 6, the model with the Residual Block achieves smoother boundary extraction and more complete watershed delineation, indicating its positive effect on segmentation continuity.

With the inclusion of the BAM, the model exhibits a further and more significant accuracy improvement. The attention mechanism enables the network to focus more precisely on key areas such as watershed boundaries, thereby enhancing boundary sensitivity and delineation accuracy. As shown in Fig. 6, the BAM-enhanced model produces finer and more closed boundary lines. However, its performance in relatively flat terrain is slightly inferior, suggesting that a single architectural modification alone cannot achieve full global stability and consistency, and that collaborative optimization with other modules remains necessary.

After incorporating the MFFM, the model continues to improve across multiple metrics. The MFFM enhances feature transmission and semantic fusion by integrating encoder and decoder features from different levels, along with the output from the previous MFFM. This design establishes cross-level semantic pathways and effectively mitigates the limited information flow inherent in the original SegNet architecture. The fusion mechanism strengthens multi-scale feature interaction, allowing high-level semantic information and low-level spatial details to be fully integrated during decoding. As observed in Fig. 6, the model with MFFM shows fewer misclassifications within watershed interiors and achieves more continuous and coherent boundary delineation. These results indicate that multi-level feature fusion substantially improves the model's capacity to perceive complex spatial structures, enabling it to capture fine-grained details.

Finally, the RBM-SegNet, which integrates the Residual, BAM, and MFFM modules, achieves the highest performance across all metrics. This confirms that the three modules form a complementary and synergistic relationship in feature extraction, attention allocation, and multi-scale information fusion. Together, they construct a closed-loop optimization framework that enables efficient feature selection and

Table 4

Performance comparison of RBM-SegNet under different coefficient settings (a_1, a_2) in the MFFM module.

Model RBM-SegNet	MFFM		Accuracy	F1-score	mIoU	FWIoU	TNR
	a_1	a_2					
1	0.5	0.5	79.41%	57.57%	48.51%	77.31%	78.83%
2	0.6	0.4	89.45%	60.47%	52.45%	87.54%	89.80%
3	0.7	0.3	92.33%	64.46%	56.26%	90.52%	94.75%
4	0.8	0.2	91.68%	64.00%	55.69%	89.86%	92.10%
5	0.9	0.1	83.53%	60.17%	51.15%	81.46%	83.08%

continuous information transmission. This mechanism effectively overcomes the limitations of the conventional SegNet, such as weakened fine-grained feature representation and insufficient contextual awareness. To further verify the effectiveness of the MFFM module and determine the optimal balance between the current-layer and cross-level feature contributions, a series of parameter sensitivity experiments were conducted by adjusting the fusion coefficients a_1 and a_2 in Eq. (1).

As presented in Table 4, different combinations of a_1 and a_2 were tested to determine the optimal balance between current-layer and cross-level feature contributions. When a_1 is set too low, the network relies excessively on cross-level features, which weakens the representation of current semantic information and leads to a marked decrease in segmentation accuracy. Conversely, when a_1 is too high, the model becomes biased toward shallow features, limiting the propagation of high-level contextual information and causing a decline in overall performance, particularly in complex topographic regions. The best performance is achieved when $a_1 = 0.7$ and $a_2 = 0.3$, where the model attains the highest accuracy. This configuration provides an optimal equilibrium between detailed local features and broad contextual semantics.

4.4. Evaluating the proposed framework's performance through case studies

This section reports the performance of the proposed method in extracting watershed boundaries across different regions and compares it with the D8 flow-routing method. The boundaries initially generated undergo a post-processing procedure, including boundary refinement, the removal of non-closed boundaries, and enhancement based on FA. The relevant workflow is illustrated in Fig. 7.

For quantitative evaluation, we selected three samples from test data to cover both geomorphological diversity and regional variation: a mountainous area and a flat area from Wisconsin, USA, and a third sample from Shaanxi, China. For the reference boundary data of Shaanxi, we redrew the boundaries at a 60-meter resolution based on the average area of the highest-level sub-basins in the 410-meter resolution HydroBASIN dataset. Because the sample represents a large area, RBM-SegNet splits the data into 512×512 pixel patches, performs individual predictions, and then stitches them together to form the final boundary map. In contrast, the D8 flow-routing and the burning stream method require multiple rounds of parameter tuning to extract the major watershed boundaries within the sample area. Table 5 summarizes

Table 5

Accuracy assessment of initial watershed boundary extraction using RBM-SegNet, D8 flow routing and burning stream method.

Models	Accuracy	F1-score	mIoU	FWIoU	TNR	Inference time/patch	Total time
D8	96.91%	59.34%	54.08%	95.23%	98.46%	0.60 s	76.00 s
D8 + burning stream	96.97%	59.35%	54.08%	95.26%	98.52%	1.30 s	197.00 s
RBM-SegNet	97.15	59.38%	54.19%	95.48%	98.71%	0.43 s	65.36 s

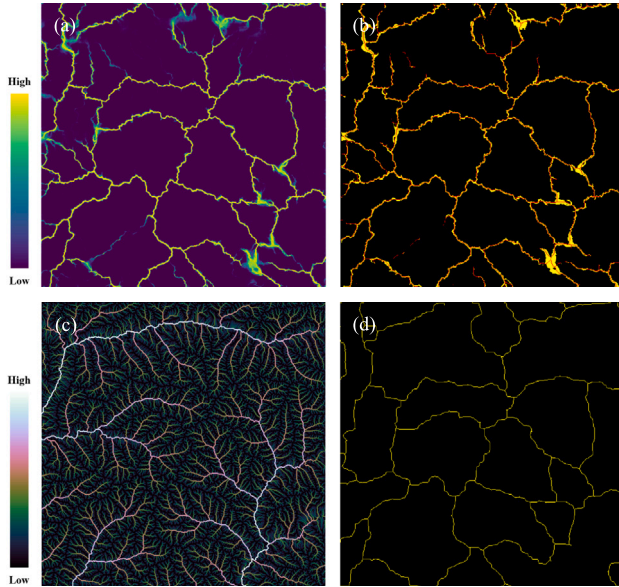


Fig. 7. Postprocessing for extracting the watershed boundary. (a): the probability map of boundaries represented by softmax values; (b): boundary skeletons obtained by thresholding predicted boundary probability and the skeletonization results, where red lines are the extracted midlines; (c): flow accumulation based on the D8 algorithm; (d): connected watershed boundary.

the accuracy metrics of the three methods relative to the reference boundaries.

To compare the methods' watershed boundary-extraction performance more intuitively Fig. 8 presents visualizations of three test regions. In both the flat and mountainous areas of Wisconsin, USA (Fig. 8(a)–(b)), overall, the watershed boundaries generated by the D8 flow-routing method appear relatively smooth and well-connected. However, since this method relies on threshold-based control mechanisms, multiple rounds of parameter tuning are often required to achieve satisfactory delineation. In particular, when clear criteria for division are lacking, noticeable misclassifications still occur, as indicated by the blue boxes. Regarding the flat terrain shown in Fig. 8(a), the D8 method often fails to determine a dominant slope direction, resulting in irregular watershed boundaries (highlighted by the purple boxes). After incorporating the stream-burning method, the overall accuracy remained nearly unchanged compared with the traditional D8 approach, with only minor improvements observed in local boundary continuity. This method optimizes flow routing and stream delineation by lowering the elevations of DEM cells affected by the river mask. However, based on the large-scale experiments in this study, the influence of stream burning on watershed boundary extraction is limited, while the computational cost increases significantly. As shown in Table 5, the average inference time per 512×512 patch increased from 0.60 s for the D8 method to 1.30 s for the stream-burning method, while the total processing time for the three case-study regions rose from 76.00 s to 197.00 s. In contrast, the RBM-SegNet model integrates the DEM and various terrain factors to learn the characteristics of boundary regions and extract appropriate watershed boundaries. It demonstrates robust adaptability even under complex terrain conditions. The experimental

results show that the semantic segmentation-based RBM-SegNet can automatically identify boundary features and generate more accurate watershed boundaries, achieving both higher accuracy and faster inference compared with the conventional algorithms.

Concerning the mountainous region of Shaanxi (Fig. 8(c)), because the threshold settings were ambiguous, the D8 method tends to over-segment the region and produce relatively small sub-watersheds. After introducing the stream-burning method, the overall delineation results still show no significant changes compared with the traditional D8 approach. Although RBM-SegNet also presents some misclassifications, its overall extraction results are more consistent with the reference boundaries. These findings indicate that RBM-SegNet maintains strong watershed extraction capability across different terrains and regions, demonstrating its potential for wide application.

Fig. 9 provides an enlarged comparison of two typical issues observed in Fig. 8. In flat regions with wide rivers, such as those shown in Fig. 9(a) and Fig. 9(b), the D8 algorithm often fails to identify the steepest dominant slope, instead assigning a predefined “default” flow direction. This leads to inaccurate flow routing and unrealistic watershed boundaries. Moreover, using a general threshold in the D8 method prevents it from accurately delineating small sub-watersheds, as shown in Fig. 9 panels (c), (d), and (e), since the threshold is primarily designed for larger basins. Although the stream-burning method locally modifies the DEM by lowering the elevations along predefined river channels to improve flow routing, its overall impact on watershed delineation remains limited. For large-scale watershed, most cells eventually converge to the same outlet, and the locally burned channels only make minor corrections to the microtopography without significantly altering the overall flow structure. In wide valley regions, where the main river channels already exhibit clear elevation gradients in the original DEM, the effect of stream burning on optimizing flow direction is relatively weak. It is noteworthy that in Fig. 9(f), the stream-burning process produces a more refined river network, resulting in a more fragmented watershed delineation with an increased number of small sub-watersheds. In contrast, the watershed boundaries extracted by RBM-SegNet effectively overcome the issues of unreasonable segmentation in flat areas and somewhat achieve delineation that aligns with the predefined criteria. This suggests that the model can effectively learn the geographic characteristics of watershed boundaries and accurately capture their spatial distribution.

5. Conclusion

This paper presents a semantic segmentation-based framework for automatically extracting watershed boundaries from Digital Elevation Models (DEMs), thereby addressing the limitations of existing methods. Traditional approaches often rely on manually set thresholds and tend to generate unreasonable boundaries in complex terrain regions. In contrast, the developed framework eliminates the need for user-defined thresholds by learning and expressing the critical features of watershed boundaries, effectively overcoming these challenges.

The developed framework comprises two key steps: the construction and training of a semantic segmentation model, and a post-processing procedure. Concerning model design, we developed the RBM-SegNet model, which builds upon the baseline SegNet architecture. RBM-SegNet introduces residual connection and the Bottleneck Attention Module (BAM) to enhance the model's ability to learn and express key information. It incorporates the Multi-Feature Fusion Module (MFFM)

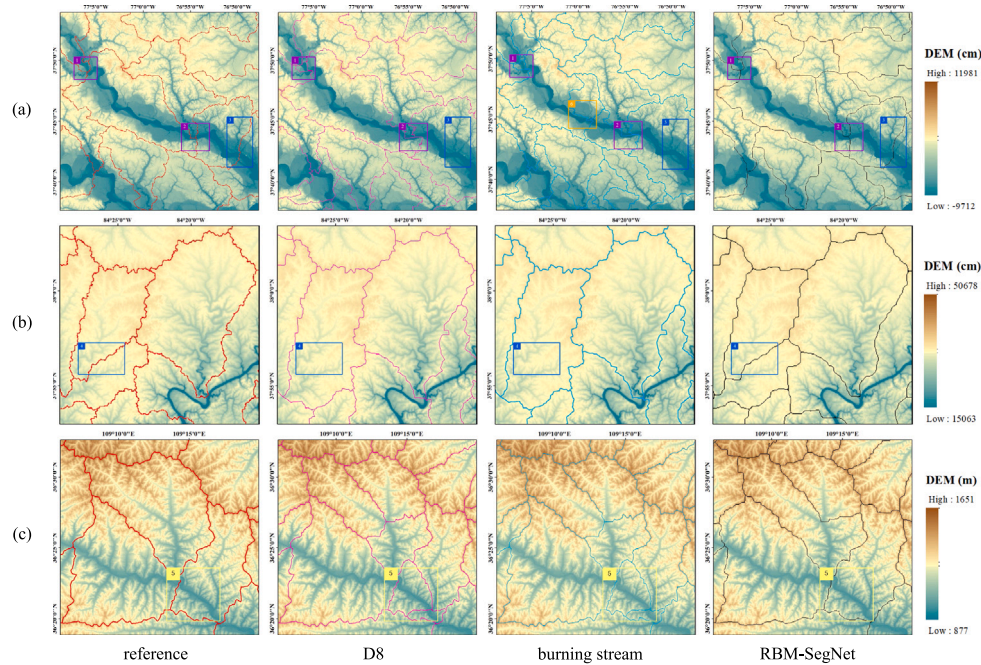


Fig. 8. Comparison of boundaries extracted by RBM-SegNet, D8 flow routing method and burning stream method with reference data in: (a) a flat region of Wisconsin, USA; (b) a mountain region of Wisconsin, USA; (c) Shannxi, China.

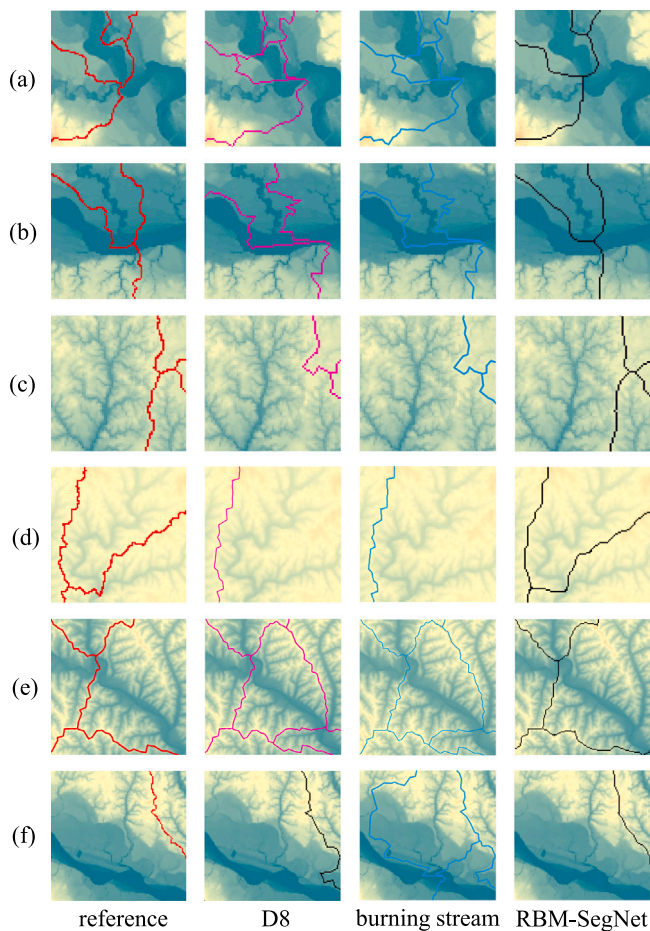


Fig. 9. Detailed enlarged view of Fig. 8.

in the decoder to mitigate the information loss during upsampling. These improvements address the shortcomings of the baseline model regarding boundary closure and spatial continuity. To enhance boundary accuracy and connectivity, a flow accumulation-based post-processing step refines and completes the extracted boundaries, making the final results more consistent with natural hydrological processes. The study's experimental results show that RBM-SegNet achieved high segmentation accuracy on the test set and can generate complete and continuous boundaries even in complex terrain conditions, overall outperforming existing deep learning models and the traditional method. In addition, this study evaluated several input combinations and finally adopted a four-channel input [DEM, Slope, HillShade, Aspect] to enhance the model's ability to perceive terrain features. The model's generalization capability was also evaluated using cross-regional datasets. However, the extracted watershed boundaries were not explicitly constrained by hydrological delineation standards, such as minimum catchment area. Since the segmentation process is driven solely by local pixel-wise features, it may produce over-segmented or incomplete watershed units in certain cases. Future work could explore a multi-task learning framework that integrates both boundary segmentation and hydrological constraints to improve accuracy and reliability in watershed delineation.

CRediT authorship contribution statement

Huanyu Yang: Writing – original draft, Visualization, Methodology, Investigation, Data curation. **Hongming Zhang:** Writing – original draft, Methodology, Data curation. **Yuwei Sun:** Investigation, Data curation. **Lu Du:** Data curation. **Weilin Xu:** Writing – review & editing, Data curation. **Jincheng Ni:** Writing – review & editing. **Qiankun Chen:** Writing – review & editing. **Chunmei Wang:** Writing – review & editing. **Qinke Yang:** Writing – review & editing. **Haijing Shi:** Writing – review & editing.

Software and data availability section

Name of software: RBM-SegNet
Developers: Huanyu Yang

Contact: yhynwafu@163.com

Date first available: 2025

Program language: Python

Source code at: <https://github.com/COCO-Y31/RBM-SegNet>

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Hongming Zhang reports financial support was provided by the National Natural Science Foundation of China (Grant No. 42377341), the Key Research and Development Program of Shaanxi (Grant No. 2023-YBNY-217), the National Key Research and Development Program of China (Grant No. 2023YFD1300101), Shenzhen Science and Technology Innovation Commission (Grant No. KCXST20221021111609022).

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Data availability

Data will be made available on request.

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